0032

National University of Computer and Emerging Sciences
Lahore Campus

Operating Systems (CS2018)

Sessional-II Exam

Date: April 7th 2026

Course Instructor(s)

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Total Time (Hrs): 1

Total Marks: 30

Total Questions: 3

Do not write below this line

Attempt all the questions.

CLO 3: Analyze memory management techniques.

Q1: Find out the number of page fault if there are three page frames, using the following page replacement algorithms: [10 Marks]

1) FIFO 3
2) Optimal 4
3) LRU 3

for the following page reference string, 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6.

CLO #: 3 Analyze memory management techniques.

Q2: Consider a swapping system in which memory consists of the following partitions (hole sizes) in memory order: 300KB, 600KB, 350KB, 200KB, 750KB and 125KB. [10 Marks]

Investigate how would the first fit (3), best fit (3) and worst fit (4) algorithm place processes of size 115KB, 500KB, 358KB, 200KB, 375KB (in order) 10

CLO #: 3 Analyze memory management techniques.

Q3a: Deduce the condition of link list after X terminates for a, b, c and d.

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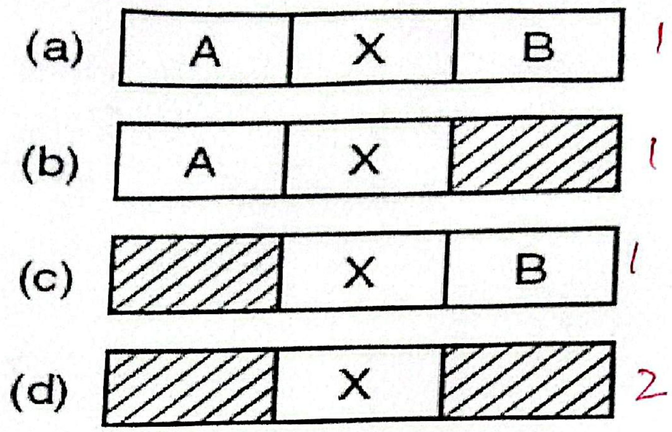

[5 Marks]

2

2

Q3b: Explain how Clock page algorithm is better than second chance algorithm using the following data. Enter Page F, G and H.

Hint: A is at head in second chance algorithm, B is second and so on. In Clock page A is at pointing arm, B is after that and so on.

1 → Explain

Page	R bit
A	1
B	0
C	1
D	1
E	0

[5 Marks]

CE(b)

Round Robin:

QT = 3sec.

Process	AT (T_0)	BT (ΔT)	Final Time (T_1)	Turn around time TAT = $T_1 - T_0$	Waiting time WAT = TAT − ΔT
P1	0	5	11	11	6
P2	1	7	19	18	11
P3	2	4	15	13	9
P4	12	6	22	10	4

Gantt chart:

P1	P2	P3	P1	P2	P3	P4	P2	P4

0 3 6 9 12 14 15 18 19 22

11

Queue:

3sec		6sec	9sec	11sec
P1	P2	P3	P1	P2
P2	P3	P1	P2	P3
P3	P1	P2	P3	

(Rear

7.5

Remaining time:

P_1 : 5 , 5−3=2 , 2−2=0 (x)

P_2 : 7 , 7−3=4 , 4−3=1

P_3 : 4 , 4−3=1 (x)

P_4 : 6 , 6−3=3

14sec		18sec
P3		P2
P4	→ new process	P4
P2	→ previous	

$$\text{Average TAT} = \frac{11+18+13+10}{4} = 13$$

$$\text{WAT} = \frac{6+11+9+4}{4} = 7.5$$